ORGNZ-03: Bucket Strategy and Design

> **Series:** ORGNZ | **Notebook:** 3 of 9 | **Created:** January 2026 | **Last Updated:** 01/28/2026

Overview

A well-defined bucket strategy optimizes query performance, controls costs, and ensures compliance. This notebook covers naming conventions, retention planning, organizational alignment, and common bucket design patterns.

Prerequisites

| Requirement | Details |
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| **Dynatrace Environment** | SaaS environment with Grail enabled |

| **Permissions** | `storage:bucket-definitions:write` for bucket creation |
| **Knowledge** | Completed ORGNZ-02 |

Learning Objectives

By the end of this notebook, you will:
- Design effective bucket naming conventions
- Plan retention strategies for different use cases
- Understand cost implications of bucket design
- Apply organizational alignment patterns

Bucket Naming Conventions

Recommended Naming Patterns

Pattern	Format	Example	Use Case
Provider-based	`<provider>_<type>_<retention>`	`aws_logs_35d`	Multi-cloud environments
LOB-based	`<provider>_<type>_<lob>`	`aws_logs_finance`	Line of business cost attribution
Team-based	`team_<name>_<type>`	`team_platform_logs`	Team ownership and billing
Compliance	`<regulation>_<type>_<retention>`	`hipaa_logs_7y`	Regulatory requirements

Detailed Examples

Bucket Name	Provider	Data Type	Retention	Org Unit
`aws_logs_35d_platform`	AWS	logs	35 days	Platform team
`azure_metrics_90d_finance`	Azure	metrics	90 days	Finance
`gcp_spans_14d_checkout`	GCP	spans	14 days	Checkout service
`audit_logs_365d_compliance`	Any	logs	365 days	Compliance
`debug_logs_7d`	Any	logs	7 days	Development

Naming Rules Reminder

Rule	Requirement
First character	Must be a letter
Allowed characters	Lowercase alphanumeric, underscores, hyphens
Case	Lowercase only
Reserved names	Cannot use `default_*` prefix
Immutability	Names cannot be changed after creation

Retention Strategy

Retention Period Guidelines

Use Case	Recommended Retention	Rationale
Debug/Development	3-7 days	Short-term troubleshooting, high volume
Standard Operations	35 days	Monthly trends, incident response
Performance Analysis	60-90 days	Quarterly comparisons
Compliance/Audit	365-3657 days	Regulatory requirements

Retention by Data Type

Data Type	Typical Retention	Notes
Debug logs	3-7 days	High volume, low long-term value
Application logs	35-90 days	Operational analysis
Audit logs	1-7 years	Compliance requirements
Metrics	35-90 days	Trending and alerting
Spans	10-35 days	APM troubleshooting
Security events	90-365 days	Security investigations

Bucket Design Patterns

Pattern 1: Long-Term Compliance Data

Store critical audit information for extended periods:

```
Bucket: compliance_audit_logs
Table: logs
Retention: 1825 days (5 years)
Use case: SOX compliance, security audits
Route via: OpenPipeline filter on audit-level logs
```

Pattern 2: Short-Term Debug Logs

Reduce costs by storing verbose logs briefly:

```
Bucket: debug_logs_7d
Table: logs
Retention: 7 days
Use case: Development troubleshooting
Route via: OpenPipeline filter on loglevel=DEBUG
```

Pattern 3: Team Isolation

Separate buckets for team-based access and cost attribution:

```
Bucket: team_platform_logs
Table: logs
Retention: 35 days
Use case: Platform team owns infrastructure logs
Access: IAM policy restricts to platform team
```

Pattern 4: Application-Specific Retention

Business units with specific retention needs:

```
Bucket: finance_app_logs
Table: logs
Retention: 180 days
Use case: Financial application compliance
Cost: Chargeback to Finance cost center
```

Cost Control and Attribution

Cost Factors

Factor	Impact on Cost
Ingest volume (GiB/day)	Direct relationship
Retention period	Storage costs increase with longer retention
Query frequency	Query costs accumulate

Cost Attribution Strategy

Organization Level	Bucket Strategy
Business Unit	One bucket per BU
Cost Center	Map buckets to cost centers
Cloud Account	Separate by AWS/Azure/GCP account
Application	Critical apps get dedicated buckets

Analyzing Bucket Usage

```
// Analyze log volume by bucket over last 24 hours
fetch logs
| summarize recordCount = count(), by:{dt.system.bucket}
| sort recordCount desc
```

```
// Track log ingest trends by bucket over time
fetch logs
| summarize recordCount = count(), by:{time_bucket = bin(timestamp, 1h),
dt.system.bucket}
| sort time_bucket desc
```

```
// Compare bucket retention configurations
fetch dt.system.buckets
| fields name, dt.system.table, retention_days
| sort retention_days desc
```

Bucket Strategy Considerations

Access Control Options

Buckets can serve as an access control mechanism:

Access Method	Description	When to Use
Security context (record-level)	Fine-grained ABAC at the record level	Complex multi-team scenarios
Bucket-level IAM policies	Restrict access to entire buckets	Simple team isolation
Combined approach	Security context + bucket policies	Maximum control and flexibility

Bucket-Based Team Isolation Example:

```
Team A: Access to team_a_logs bucket only
Team B: Access to team_b_logs bucket only
Platform: Access to all buckets
```

Query Performance at Scale

Bucket sizing directly impacts queryability due to the 500 GB scan limit:

Bucket Ingest	Impact
~1 TB/day	Optimal - can query ~12 hours of data
1-3 TB/day	Acceptable - limited query window
>3 TB/day	Split bucket or use aggregations

Data Immutability

Operation	Possible?	Alternative
Move data between buckets	No	Route new data via OpenPipeline
Selective record deletion	No	Wait for retention to expire
Bucket deletion	Yes	Deletes ALL data permanently

Routing Data to Buckets

Use OpenPipeline to route data to custom buckets:

```
# OpenPipeline routing rule example
processors:
  - type: route
    rules:
      - condition: "loglevel == 'DEBUG'"
        destination: "debug_logs_7d"
      - condition: "log.source contains 'audit'"
        destination: "audit_logs_365d"
      - condition: "host.group starts-with 'finance-'"
        destination: "finance_app_logs"
    default: "default_logs"
```

Bucket Design Checklist

Use this checklist when planning buckets:

- [ ] Defined naming convention aligned with organization
- [ ] Identified retention requirements by data type
- [ ] Mapped organizational units to buckets for cost attribution
- [ ] Calculated expected ingest volumes (keep <1 TB/day)
- [ ] Planned for compliance requirements
- [ ] Documented bucket purposes
- [ ] Planned OpenPipeline routing rules
- [ ] Considered access control strategy (bucket vs security context)

Next Steps

Continue with the ORGNZ series:
- **ORGNZ-04**: Permissions in Grail Overview

References

- Grail Buckets
- Data Retention
- OpenPipeline Routing
